The Problem

Modern software is developed at unprecedented speed — but security testing continues to lag behind, often deferred to the final stages of the development lifecycle. This delay creates compounding risks that are expensive and difficult to reverse.

The Root Cause

Organisations rely heavily on **external penetration testing** — an approach that is infrequent, costly, and fundamentally reactive.

- Increased security costs
- Delayed release cycles
- Limited continuous visibility
- Weak audit readiness

The Result

Vulnerabilities surface late in the cycle — when fixing them is **significantly more expensive** and operationally disruptive. The later a flaw is found, the higher the remediation cost, the greater the risk, and the harder it becomes to maintain stakeholder confidence.

Key Insight: Security must be integrated earlier and continuously throughout the development lifecycle — not bolted on at the end.

Our Solution

INTRODUCING SCANITYCHECK

ScanityCheck is an **internal vulnerability scanning platform** purpose-built to continuously assess web applications and APIs throughout the software development lifecycle — shifting security left where it has the most impact.

Detect Early

Identify vulnerabilities before they reach production, reducing downstream risk and remediation cost.

Automate Testing

Replace manual, one-off assessments with automated, repeatable scanning workflows that run continuously.

Audit-Ready Reports

Generate structured, exportable reports that satisfy compliance requirements and auditor expectations.

Strengthen Practices

Embed security culture into everyday development — not just during formal audit periods.

Value Proposition: A unified platform that transforms security testing from a one-time activity into a continuous, integrated process.

Why It Matters

ScanityCheck delivers measurable value across every stakeholder group — aligning security goals with business, development, and compliance outcomes.

For Businesses

- Reduced security expenditure through continuous scanning vs. costly periodic pentests
- Lower risk exposure and faster breach prevention

For Developers

- Faster, real-time feedback integrated into their workflow
- Clearer remediation guidance to fix issues efficiently

For Security Teams

- Continuous, automated testing across all targets
- Centralised visibility into the organisation's vulnerability posture

For Compliance & Auditors

- Structured, timestamped evidence collection
- Reliable reporting that supports audit and governance requirements

Core Capabilities

ScanityCheck is built around a comprehensive feature set that covers the full vulnerability management lifecycle — from detection through evidence collection to reporting and historical analysis.

7 Tools
Integrated Scanners

4 Roles
Access Levels Tested

3 Formats
Export Options

Black-Box Scanning

Simulate external attacker behavior against web and API targets without source code access.

Authenticated API Testing

Test APIs with valid credentials to uncover deeper, access-dependent vulnerabilities.

Role-Based Access Testing

Evaluate security across Admin, Manager, and Staff privilege levels to catch access control gaps.

Severity Classification

Prioritise findings automatically from Critical down to Informational for efficient triage.

Evidence Logging

Capture full request/response data for every finding — building an irrefutable audit trail.

Exportable Reports

Generate professional reports in CSV and PDF formats ready for stakeholders and auditors.

Historical Trend Analysis

Track vulnerability remediation over time and measure your improving security posture.

Key Advantage: Detection, evidence, and reporting — unified in one integrated platform.

Security Coverage

OWASP-ALIGNED

ScanityCheck targets the vulnerabilities that matter most in production environments — drawn directly from industry-standard threat frameworks and real-world attack patterns. We analyse actual attack surfaces, not just surface-level checks.

🔐 Authentication & Access

- Broken Authentication
- Broken Access Control
- Role-privilege escalation

💉 Injection & Input

- SQL, Command, LDAP Injection
- Cross-site scripting (XSS)
- OWASP Top 10 (Web & API)

⚙️ Configuration & Headers

- Security misconfigurations
- Missing security headers
- Exposed sensitive endpoints

🔗 API-Specific Risks

- API vulnerabilities (OWASP API Top 10)
- Rate limiting bypass
- Improper asset management

How It Works

ScanityCheck follows a structured, end-to-end workflow — from target registration through to actionable reporting and long-term trend tracking. Each step is automated to minimise manual effort and maximise consistency.

Register Target

Trigger Automated Scan

Store Findings with Evidence

Generate Reports

Select Scanning Method

Analyse Vulnerabilities

Classify Severity

Track History & Trends

Built on a modern, scalable architecture: **.NET Web API** backend, **React** frontend, **SQL Server** for persistent storage, and **Hangfire** for reliable background job processing — ensuring scans run efficiently without blocking the user interface.

System Architecture

HOW SCANITYCHECK PROCESSES SCANS

From CVE intelligence to your dashboard — ScanityCheck automates the entire vulnerability detection pipeline with zero manual handoffs.

NVD
CVE threat intelligence feed

Scan Engine
Core processing & job dispatch

Scanning Devices
Web apps, APIs & servers

Scan Database
Evidence & findings storage

Scan Results
Severity-classified findings

Report Generation
PDF & CSV output

User GUI
Live dashboard & exports

NVD

Threat Intelligence Source

7 Steps

Automated Pipeline

Real-Time

Dashboard Updates

Integrated Security Tooling

Rather than forcing security teams to manage a fragmented collection of standalone tools, ScanityCheck consolidates industry-leading open-source scanners into a **single, unified workflow** — with results aggregated, deduplicated, and presented in one place.

Dynamic & API Testing

OWASP ZAP

Industry-standard web app scanner for dynamic analysis

Nuclei

Fast, template-based vulnerability detection at scale

RESTler

Stateful REST API fuzzing and security testing

Schemathesis

Schema-driven API testing for OpenAPI and GraphQL

Static & Dependency Analysis

Semgrep

Lightweight static analysis for code-level vulnerability detection

CodeQL

Semantic code analysis for deep vulnerability discovery

OWASP Dependency-Check

Identifies known vulnerabilities in third-party libraries

Trivy

Container and infrastructure vulnerability scanning

Key Benefit: One platform, many tools — eliminate fragmentation and unify all security results in a single pane of glass.

System Demonstration

The ScanityCheck platform translates complex, raw scanner output into a **clear, actionable dashboard** — designed for both security engineers and executive stakeholders who need security insight without technical overhead.

Target Overview

Manage all registered applications in one place with live status indicators and scan history at a glance.

Real-Time Scan Status

Monitor active scans as they progress, with live updates on discovery, classification, and completion.

Severity Breakdown

Visual severity tiers from Critical to Informational — enabling immediate triage prioritisation.

Report Export

One-click PDF and CSV export for audit submissions, stakeholder briefings, and compliance documentation.

Historical Trend Analysis

Track remediation velocity and vulnerability trends over time to measure security program effectiveness.

Traditional vs. ScanityCheck

FROM REACTIVE TO PROACTIVE

The shift from **reactive, periodic security assessments** to a **proactive, continuous model** is not just a technical upgrade — it's a fundamental transformation in how organisations manage risk. ScanityCheck makes that transformation operationally achievable.

Compliance & Governance

Security without structure is just noise. ScanityCheck is designed from the ground up to support organisational governance frameworks — providing the evidence, transparency, and reporting infrastructure that modern compliance demands.

Secure SDLC

Integrates security testing directly into the software development lifecycle, ensuring vulnerability management is continuous rather than episodic.

Audit-Ready Evidence

Every scan generates timestamped findings with full request/response logs — providing irrefutable, structured evidence for internal and external audits.

ISO 27001 Readiness

Supports key ISO 27001 controls around vulnerability management, risk treatment, and continuous monitoring — accelerating certification readiness.

Governance Reporting

Exportable reports structured for compliance submissions, board-level risk reviews, and regulatory engagement.

Future Enhancements

PRODUCT ROADMAP

ScanityCheck's current capabilities represent a strong foundation. The roadmap ahead focuses on deeper integration, intelligent automation, and evolving the platform into a fully embedded **DevSecOps solution** that lives within the developer workflow — not alongside it.

CI/CD Pipeline Integration

Embed automated security scans directly into build and deployment pipelines — blocking vulnerable code before it ships.

Automated Jira Ticket Generation

Automatically create, assign, and prioritise remediation tickets in Jira when critical vulnerabilities are detected.

Advanced Analytics Dashboards

Rich visual reporting on vulnerability trends, MTTR metrics, and team-level security performance over time.

AI-Driven Vulnerability Prioritisation

Machine learning models that rank findings by exploitability, business impact, and remediation urgency — cutting through noise.

Enhanced API Fuzzing

Deeper, context-aware fuzzing capabilities targeting complex stateful API interactions and edge-case attack vectors.

Closing: Security, Built In — Not Bolted On

The era of treating security as an afterthought is over. ScanityCheck empowers organisations to make **continuous, intelligent security** a native part of how software is built, tested, and deployed.

🔍 Detect Earlier

Find vulnerabilities during development — not after deployment — dramatically reducing remediation cost and impact.

⚡ Fix Faster

Automated workflows, severity triage, and evidence logs help teams resolve issues with speed and precision.

✅ Stay Compliant

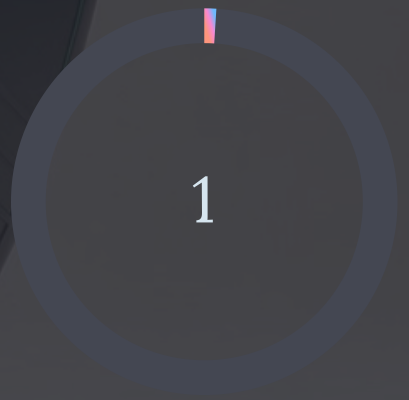
Audit-ready reports and continuous evidence collection keep your organisation prepared for any regulatory review.

🏗️ Build Securely

Embed security into the DNA of your development culture — making every release more trustworthy than the last.

Cheaper to fix early vs. late

Continuous Coverage



Unified Platform

> **"ScanityCheck brings security into everyday development — not just audits."**